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RESEARCH VISION

I work on **trustworthy AI systems built on large language models**. Today's LLMs read untrusted content and act through tools, so trusting them takes more than a better filter. My research brings **LLM verification** and **LLM security** together: checking what these systems produce, understanding how they fail under attack, and recognizing when they should refuse.

- **Current Focus: Verification and Security of LLM Systems**

Linköping University, RESIST

As a postdoctoral researcher in RESIST, I study how LLM-based systems can be made reliable and secure. I aim to express safety and correctness requirements as **properties that can be checked** on model outputs, to measure how these systems behave under **adversarial inputs**, and to design **refusal mechanisms** and transparent specifications that support external auditing.

- **Foundation: Formal Language Theory Meets Modern AI**

Ph.D., Yonsei University

My Ph.D. research connected formal language theory with AI and NLP. I studied foundational questions such as the decidability of string properties (Parikh matrices), language hierarchies, and formal characterizations of structural constraints. I then applied these tools to **verifiable generation and program analysis**, including regex transformation checked by automata equivalence, contract-based evaluation of generated code, and time-complexity prediction for code.

CURRENT POSITION

Linköping University

Linköping, SE

Postdoctoral Researcher, Department of Computer and Information Science (IDA)

2026.09–present

- Trustworthy Systems Group (TSG), Division of Cybersecurity (CYBER)
- Part of RESIST (Resilience and Security for Trustworthy AI Systems), working with Simin Nadjm-Tehrani

EDUCATION

Yonsei University

Seoul, KR

Integrated Ph.D. course in Computer Science

2019.03–2026.02

Supervisor: Yo-Sub Han

Thesis: *Symbolic Techniques for Deep Learning in Natural Language Processing and Program Analysis*

Yonsei University

Seoul, KR

Bachelor's degree in Computer Science

2015.03–2019.02

PUBLICATIONS

Full list: [\[Google Scholar\]](#) [\[DBLP\]](#)

† indicates equal contribution

Peer-Reviewed Conference/Journal Papers

- [C1] **ContractEval: A Benchmark for Evaluating Contract-Satisfying Assertions in Code Generation**
Soohan Lim[†], Jay (Joonghyuk) Hahn[†], Hyunwoo Park, Sang-Ki Ko, Yo-Sub Han
In Findings of ACL 2026, 42549–42566 [\[PDF\]](#)

- [C2] **Query4Regex: Verifiable Regex Transformation through Formal Operations from NL and DSL Queries**
Jay (Joonghyuk) Hahn, Yo-Sub Han
In Findings of EACL 2026 [PDF]
- [C3] **Repairing Regex Vulnerabilities via Localization-Guided Instructions**
Sicheol Sung[†], Jay (Joonghyuk) Hahn[†], Yo-Sub Han
In Proceedings of EACL 2026 [PDF]
- [C4] **A Regex Minimization Benchmark: A PSPACE-Complete Challenge for Language Models**
Hyundong Jin, Jay (Joonghyuk) Hahn, Yo-Sub Han
In Proceedings of EACL 2026 [PDF]
- [C5] **EnCur: Curriculum-Based In-Context Learning with Structural Encoding for Code Time Complexity Prediction**
Jay (Joonghyuk) Hahn, Aditi, Seung-Yeop Baik, Shinwoo Park, Sang-Ki Ko, Yo-Sub Han
In ESWA 2026 [PDF]
- [C6] **AmpleHate: Amplifying the Attention for Versatile Implicit Hate Detection**
Yejin Lee, Jay (Joonghyuk) Hahn, Hyeseon Ahn, Yo-Sub Han
In Proceedings of EMNLP 2025 [PDF]
- [C7] **CodeComplex: Dataset for Worst-Case Time Complexity Prediction**
Seung-Yeop Baik, Jay (Joonghyuk) Hahn, Jungin Kim, Mingi Jeon, Aditi, Yo-Sub Han, Sang-Ki Ko
In Findings of EMNLP 2025 [PDF]
- [C8] **TCPProF: Time-Complexity Prediction SSL Framework**
Jay (Joonghyuk) Hahn, Hyeseon Ahn, Jungin Kim, Soohan Lim, Yo-Sub Han
In Proceedings of NAACL 2025 [PDF]
- [C9] **Advanced Code Time Complexity Prediction Approach Using Contrastive Learning**
Shinwoo Park, Jay (Joonghyuk) Hahn, Elizabeth Orwig, Sang-Ki Ko, Yo-Sub Han
In EAAI 2025, Vol. 151 [PDF]
- [C10] **Characterizations of M-equivalence and weak M-relation**
Jay (Joonghyuk) Hahn, Hyunjoon Cheon, Yo-Sub Han
In IJFCS 2025 [PDF]
- [C11] **On the Decidability of Infix Inclusion Problem**
Hyunjoon Cheon[†], Jay (Joonghyuk) Hahn[†], Yo-Sub Han
In Theory of Computing Systems 2024, Vol. 68, 301–321 [PDF]
- [C12] **Universal Rewriting Rules for the Parikh Matrix Injectivity Problem**
Ingyu Baek[†], Jay (Joonghyuk) Hahn[†], Yo-Sub Han, Kai Salomaa
In Proceedings of DLT 2024, LNCS 14791, 68–81 [PDF]
- [C13] **SuperST: Superficial Self-Training for Few-Shot Text Classification**
Ju-Hyoung Lee[†], Jay (Joonghyuk) Hahn[†], Hyeon-Tae Seo, Jiho Park, Yo-Sub Han
In Proceedings of LREC-COLING 2024 [PDF]
- [C14] **ATHENA: Mathematical Reasoning with Thought Expansion**
J.B. Kim, Hazel Kim, Jay (Joonghyuk) Hahn, Yo-Sub Han
In Proceedings of EMNLP 2023, 16315–16327 [PDF]
- [C15] **GDA: Grammar-based Data Augmentation for Text Classification using Slot Information**
Jay (Joonghyuk) Hahn, Hyunjoon Cheon, Elizabeth G. Orwig, Su-Hyeon Kim, Sang-Ki Ko, Yo-Sub Han
In Findings of EMNLP 2023 [PDF]
- [C16] **M-equivalence of Parikh Matrix over a Ternary Alphabet**
Jay (Joonghyuk) Hahn, Hyunjoon Cheon, Yo-Sub Han
In Proceedings of CIAA 2023 [PDF]
- [C17] **Boosting Code Summarization by Embedding Code Structures**
Jikyeong Son[†], Jay (Joonghyuk) Hahn[†], Hyeon-Tae Seo, Yo-Sub Han
In Proceedings of COLING 2022, 5966–5977 [PDF]

- [C18] **On the Decidability of Infix Inclusion Problem**
 Hyunjoon Cheon[†], **Jay (Joonghyuk) Hahn[†]**, Yo-Sub Han
In Proceedings of DLT 2022, LNCS 13257, 115-126 [PDF]
- [C19] **Self-Training using Rules of Grammar for Few-Shot NLU**
Jay (Joonghyuk) Hahn, Hyunjoon Cheon, Kyuyeol Han, Cheongjae Lee, Junseok Kim, Yo-Sub Han
In Findings of EMNLP 2021 [PDF]
- [C20] **Most Pseudo-copy Languages Are Not Context-Free**
 Hyunjoon Cheon[†], **Jay (Joonghyuk) Hahn[†]**, Yo-Sub Han, Sang-Ki Ko
In Proceedings of COCOON 2021, 189-200 [PDF]

Preprints and Manuscripts under Review

- **MEC³O: Multi-Expert Consensus for Code Time Complexity Prediction**
Jay (Joonghyuk) Hahn[†], Soohan Lim[†], Yo-Sub Han
Preprint, arXiv:2510.09049 [PDF]
- **ECO: Enhanced Code Optimization via Performance-Aware Prompting for Code-LLMs**
 Su-Hyeon Kim, **Jay (Joonghyuk) Hahn**, Sooyung Cha, Yo-Sub Han
Preprint, arXiv:2510.10517 [PDF]
- **A Voronoi-Embedding Framework for Semi-Supervised Time-Complexity Prediction**
 Hyeseon Ahn, Jay (Joonghyuk) Hahn, Yo-Sub Han
Under review
- **Universal Rewriting Rules for the Parikh Matrix Injectivity Problem (Journal Extension)**
 Ingyu Baek[†], Jay (Joonghyuk) Hahn[†], Yo-Sub Han, Kai Salomaa
Under review

PROJECTS

2023–2025: Human–AI Programming Platform Research (Follow-up)

Ministry of Science and ICT (MSIT), Republic of Korea

- Studied Human-in-the-Loop workflows using large language models (LLMs) to infer developers’ programming intent and generate reliable source code.
- Investigated prompt engineering techniques to control hallucinations during code analysis and improve consistency of LLM-assisted programming.

2023–2024: Deep Learning-based Financial Data Analysis & Transformation

Bankware Global

- Led a data-engineering methodology to analyze specialized financial data and transform it into LLM-friendly formats for downstream reasoning and retrieval.
- Explored domain adaptation directions for finance-specialized models using limited expert data, including instruction tuning and retrieval-augmented generation (RAG) to improve performance.

2022: 5th AI Grand Challenge

Ministry of Science and ICT (MSIT), Republic of Korea

- Developed deep learning models for inferring solution processes and final answers for descriptive (free-form) math problems; participated in the national-scale competition finals.
- Designed equation-based heuristics to generate Korean training data; fine-tuned language models for mathematical problem solving and code-like reasoning.
- Passed the first stage and drafted the project proposal; then participated in the second stage.

2019–2022: Human–AI Collaborative Programming Platform Research

Ministry of Science and ICT (MSIT), Republic of Korea

- Designed and studied platform frameworks for efficient collaboration between humans and AI in programming workflows.

- Developed neuro-symbolic and semi-supervised learning ideas for reliable code-related tasks (e.g., code summarization and complexity-oriented analysis) under limited supervision.

2019–2022: Automated Malware Pattern Rule-set Generation

Ministry of Science and ICT (MSIT), Republic of Korea

- Developed automated pipelines to generate malware pattern rule-sets by training deep learning models for malware pattern detection.
- Studied rule-set compression and synthesis from learned signals to support scalable detection and automation.

RESEARCH INTERESTS FOR FUTURE WORKS

- **Mechanistic Interpretability of Generative Models:** Applying formal language theory and automata to understand internal mechanisms of LLMs—analyzing learned circuits, feature representations, and reasoning processes to make models more transparent and interpretable.
- **Formal Reasoning & Symbolic Integration:** Bridging symbolic methods (formal grammars, logical constraints, decidability analysis) with neural models to enable verifiable reasoning, compositional generalization, and systematic problem-solving in language models.
- **Constrained & Controllable Generation:** Developing grammar-based frameworks for structured, safe generation with guarantees on output properties—applicable to code synthesis, mathematical reasoning, and domain-specific language generation.
- **Efficient & Robust Language Models:** Investigating byte-level models, robustness to perturbations, and efficient architectures informed by formal language hierarchies—advancing foundations for more resilient and efficient generative systems.
- **Neuro-Symbolic AI Safety:** Bridging symbolic reasoning (formal methods, logical constraints) with neural models to create interpretable, verifiable AI systems with transparent failure modes.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, Java
- **ML/NLP Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers
- **Formal Methods:** Automata theory, formal grammars, decidability analysis, formal language hierarchies
- **Research Areas:** constrained generation, neuro-symbolic AI, formal verification for neural networks
- **Tools:** $\text{\LaTeX}$, Git, Linux/Unix